Synthetic Latent Diffusion Data for Low-Data IMS Classification:
An Asilomar 2026 Study

Kevin J. Metzler, Hy Lam, Cate Dunham, Randy Paffenroth
Worcester Polytechnic Institute, Worcester, MA, USA
kjmetzler@wpi.edu

Abstract—In low-data ion mobility spectrometry (IMS), classifier performance is often limited by scarce labeled spectra. We study whether synthetic samples generated by a separation-guided latent diffusion model improve untuned downstream classification. Using 8 IMS chemicals, we evaluate Random Forest and MLP classifiers trained with varying real/synthetic mixtures across 50 low-data regimes per synthetic setting. Synthetic data provides consistent but modest gains for Random Forest (mean best-minus-real: +0.0061 to +0.0097 absolute accuracy), and large gains for MLP when synthetic variance is increased (std=1.5: +0.1519, std=2.0: +0.1304). At std=1.5, the untuned MLP improves over real-only training in 49/50 regimes, with synthetic-only training outperforming real-only in 46/50 regimes. These results show that manifold-aware synthetic generation can materially improve low-data classifier training, especially for high-capacity neural classifiers.

Index Terms—ion mobility spectrometry, synthetic data, diffusion models, data augmentation, classification

I. Introduction

Ion mobility spectrometry is widely used for rapid chemical detection, but obtaining large, balanced labeled datasets remains expensive and operationally constrained [1], [2]. In this setting, synthetic data is attractive, but only if generated samples preserve discriminative structure relevant for downstream tasks.

Prior work in this project introduced a decoupled autoencoder with ChemNet-informed latent representations and a separation-guided diffusion process for synthetic spectrum generation [3]–[5]. The present paper shifts focus from generative methodology to classifier utility: does this synthetic data improve low-data classification when no classifier hyperparameter tuning is performed?

Our central finding is that synthetic augmentation is highly beneficial for untuned MLP classifiers and mildly beneficial for Random Forest, with gains depending strongly on synthetic sampling variance.

Contributions. This paper contributes: (1) a controlled low-data benchmark for real/synthetic mixing across 8 IMS chemicals, (2) quantitative evidence that synthetic augmentation produces large neural-classifier gains in untuned settings, and (3) a practical operating-point analysis showing where synthetic variance and mixing ratio help most.

II. Experimental Design

A. Data and Generative Source

We use the same 8-chemical IMS setting as the methodology study: DEB, DEM, DMMP, DPM, DtBP, JP8, MES, and TEPO. The base dataset contains 222,519 training and 74,173 test spectra. Synthetic spectra are generated from the latent diffusion pipeline at three sampling scales: $\sigma \in \{1.0, 1.5, 2.0\}$. For each scale, we generate 500 samples per class (4000 total), then evaluate classifier behavior under constrained data budgets.

B. Classifier Evaluation Protocol

For each σ, we evaluate two untuned classifiers:

- Random Forest (100 trees, default feature handling)
- MLP (architecture: 1000-500-250-100, adaptive Adam, standardized inputs)

For each class, we vary training size from 1 to 50 points/class and vary the real-data fraction $r \in \{0.0, 0.1, \ldots, 1.0\}$. Here, $r = 1.0$ is real-only training and $r = 0.0$ is synthetic-only training. For each regime (fixed points/class), we compare:

$$\Delta = \max_{r\in[0,1]} \mathrm{Acc}(r) - \mathrm{Acc}(r = 1.0) \quad (1)$$

where positive Δ indicates that adding synthetic data beats real-only training.

C. Reproducibility Notes

All results in this paper come from the project evaluation pipeline in scripts/4f-kjm-evaluatesynthetic-data.py and are summarized in results/eval_synthetic_metrics_summary.csv. We report absolute accuracy deltas (not relative percent gain), and all classifier settings are intentionally untuned to isolate augmentation effects.

III. Results

A. Overall Benefit Versus Real-Only Baseline

Table I summarizes 50 low-data regimes per classifier and synthetic setting.

Two trends are clear.

First, Random Forest gains are present but small: roughly 0.6%–1.0% absolute accuracy on average. This

TABLE I
Synthetic augmentation benefit relative to real-only training (50 regimes per row).

Classifier	Synthetic Std	Mean Δ	Median Δ	Improved Regimes	Synthetic-Only Better
Random Forest	1.0	+0.0097	+0.0000	20/50	2/50
Random Forest	1.5	+0.0061	+0.0017	30/50	0/50
Random Forest	2.0	+0.0080	+0.0006	28/50	1/50
MLP	1.0	+0.0129	+0.0011	25/50	2/50
MLP	1.5	+0.1519	+0.1618	49/50	46/50
MLP	2.0	+0.1304	+0.1214	50/50	31/50

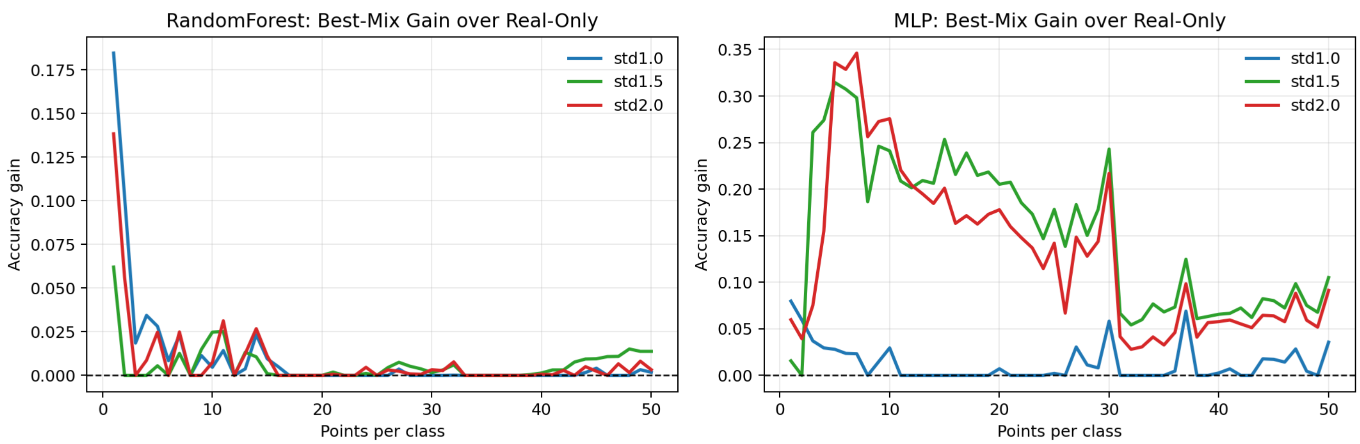

Fig. 1. Per-budget improvement over real-only training, defined as $\max_r \mathrm{Acc}(r) - \mathrm{Acc}(r{=}1.0)$ for each points-per-class budget. MLP gains are consistently large for std=1.5 and std=2.0, while Random Forest gains are positive but comparatively small.

suggests synthetic data helps coverage but RF is comparatively less sensitive to manifold expansion.

Second, MLP benefits are substantial at higher synthetic variance. At std=1.5, the average best-minus-real gain is +0.1519 (15.19 percentage points), and real-only is surpassed in 49/50 regimes. At std=2.0, gains remain large (+13.04 percentage points on average), with improvement in all 50/50 regimes.

Figure 1 adds budget-level granularity: MLP gains are strongest in the low-data regime and taper as more real data is available, while Random Forest shows smaller but stable positive deltas.

B. Budget-Stratified Analysis

To better characterize where augmentation helps, Table II reports mean gains in three budget bins.

The budget-stratified view shows that synthetic augmentation provides the largest marginal value when real data is scarcest, particularly for MLP. Even in higher-data bins, MLP retains non-trivial gains for std=1.5 and std=2.0.

C. Interpretation

The strongest gains occur for the higher-capacity classifier (MLP), consistent with the hypothesis that neural models can exploit additional manifold-supported variation from synthetic samples. The weaker RF gains suggest tree ensembles already saturate much of the low-data signal with smaller marginal benefit from synthetic diversity.

Across both models, std=1.5 appears to provide the best trade-off: enough spread to improve class support, without the larger distribution shift risk implied by more aggressive sampling.

IV. Discussion

These results indicate that synthetic data utility is not uniform across downstream models. For deployment-focused workflows:

- If using untuned MLPs in low-data regimes, synthetic augmentation is high-impact and often dominant.
- If using untuned Random Forests, synthetic data provides modest but consistent gains and can still be useful when every percentage point matters.
- Sampling scale should be treated as a first-class hyperparameter; std=1.5 was the strongest setting in this study.

Figure 2 operationalizes this guidance by showing how the best real/synthetic ratio shifts with available data budget. For neural models, synthetic-heavy training can be optimal at low n, while more real-heavy mixing becomes preferable as n increases.

TABLE II
Mean best-minus-real gain (Δ) by data budget bin.

Classifier	Synthetic Std	$n \leq 10$	$11 \leq n \leq 30$	$31 \leq n \leq 50$
Random Forest	1.0	+0.0415	+0.0030	+0.0005
Random Forest	1.5	+0.0120	+0.0038	+0.0054
Random Forest	2.0	+0.0260	+0.0049	+0.0021
MLP	1.0	+0.0325	+0.0059	+0.0101
MLP	1.5	+0.2144	+0.1978	+0.0748
MLP	2.0	+0.2144	+0.1630	+0.0558

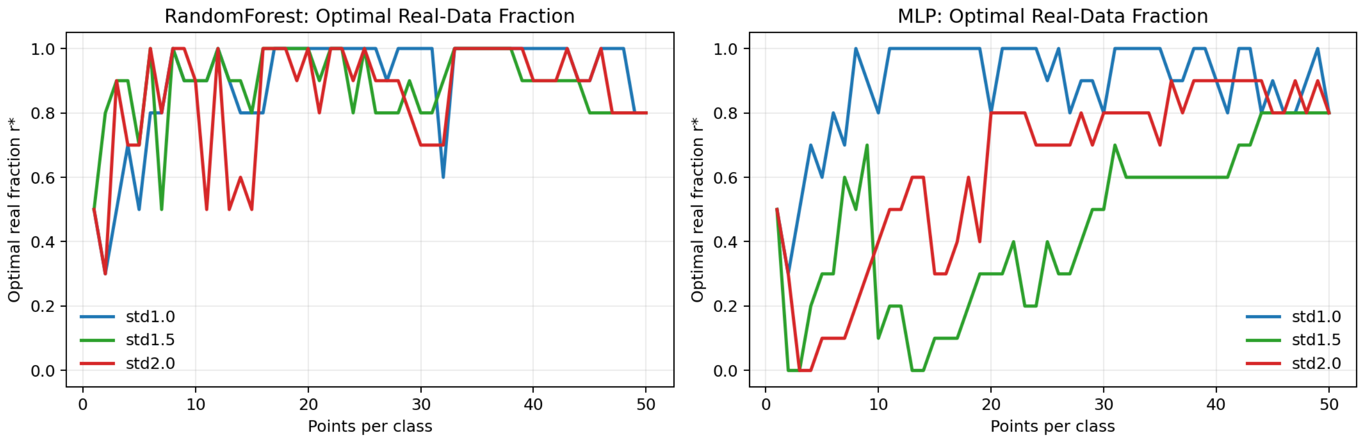

Fig. 2. Optimal real-data fraction r^* (the ratio that maximizes accuracy for each budget). For MLP at std=1.5 and std=2.0, optimal training often uses substantial synthetic content in low-data settings, while Random Forest remains closer to mixed or real-leaning ratios.

A practical implication is that generative-model evaluation should include downstream classifier lift, not only moment matching or visual plausibility.

V. Conclusion

This Asilomar-focused study demonstrates that separation-guided latent diffusion data materially improves low-data IMS classification, particularly for untuned MLPs. Relative to real-only training, synthetic augmentation yields large and consistent gains at appropriate sampling variance (best at std=1.5), while still offering modest improvements for untuned Random Forest.

Future work will extend this analysis to calibrated uncertainty-aware augmentation and tuned classifier baselines to quantify how much additional uplift remains after model selection.

References

[1] H. H. Hill Jr, W. F. Siems, and R. H. St. Louis, "Ion mobility spectrometry," Analytical Chemistry, vol. 62, no. 23, pp. 1201A–1209A, 1990.

[2] J. N. Dodds and E. S. Baker, "Ion mobility spectrometry: fundamental concepts, instrumentation, applications, and the road ahead," Journal of the American Society for Mass Spectrometry, vol. 30, no. 11, pp. 2185–2195, 2019.

[3] C. Dunham, M. Barger, R. Paffenroth, J. Uzarski, and C.-W. Tsai, "Oracle embeddings for chemical detection," in 2024 International Conference on Machine Learning and Applications (ICMLA). IEEE, 2024, to appear. [Online]. Available: https://conferences.computer.org/icmlapub24/pdfs/ICMLA2024-1MSqUZZS0oyBeWxCwySuYM/748800a272/748800a272.pdf

[4] K. Preuer, P. Renz, T. Unterthiner, S. Hochreiter, and G. Klambauer, "Fréchet chemnet distance: A metric for generative models for molecules in drug design—supporting information—," Deep Learning in Drug Discovery, p. 59, 2019.

[5] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," Advances in neural information processing systems, vol. 33, pp. 6840–6851, 2020.